CoreOps: Week 3 Module

Kubernetes Security & Platform Hardening

Objective: Master Kubernetes security fundamentals, pod-level & node-level isolation, RBAC correctness, secrets management, and supply-chain trust; the backbone of production-grade clusters.

MONDAY: Pod Security, Seccomp & AppArmor

What You Should Understand Today

- Why Kubernetes needs pod-level security
- What Pod Security Admission (PSA) is
- What "privileged vs unprivileged" workloads mean
- Intro to Seccomp & AppArmor (just high-level awareness)

Key Outcomes

- You know what Kubernetes *blocks* for safety
- You understand which fields in a pod make it "unsafe"
- You can read PSA violation messages without fear

WHAT TO PRACTICE

Lab 1: Run a Restricted Pod

Deploy a simple Nginx deployment and ensure it passes PSA restricted mode.

Lab 2: Make a Pod Break PSA

Add privileged = true and see how Kubernetes rejects it.

Lab 3: Fix the Pod

Remove unsafe fields and reapply.

WHERE TO STUDY

- Pod Security Admission (official docs)
- Basic securityContext examples

INFRATHRONE NOTES

- Most attacks start because pods run with unnecessary permissions.
- PSA is your *first layer of defence.*